9. Monday, July 8th. More on metric spaces

Last Time: · Metric Spaces (X,d) where $d: X\times X \to [0,\infty)$ non-deg. symmetric, Δ inequality.

· Metric Topology $\mathcal{T}_d = \{U \subseteq X \mid \forall x\in U\ \exists \varepsilon>0 \text{ s.t. } B(x,\varepsilon)\subseteq U\}$.

· Balls $\{B(x,\varepsilon) \mid x\in X,\ \varepsilon>0\}$ form a basis.

· On $\mathbb{R}^n$, for $p\geq 1$ metric $d_p(x,y) = \left(\sum_{i=1}^{n} |x_i-y_i|^p\right)^{1/p}$, $d_\infty(x,y) = \max_{1\leq i\leq n}\{|x_i-y_i|\}$

Examples: 1) Let X set with discrete metric $d(x,y) = \begin{cases} 0 & \text{if } x=y \\ 1 & \text{else} \end{cases}$

Then $\mathcal{T}_d = \mathcal{P}(X)$ is the discrete topology.

It is enough to show that all singletons are open in $\mathcal{T}_d$.

Given $x\in X$, we have $B(x,\frac{1}{2}) = \{x\}$, Hence $\{x\}$ open.

2) On $\mathbb{R}$ with the metric $d(x,y) = |x-y|$, the metric topology $\mathcal{T}_d$ coincides with the standard topology.

3) On $\mathbb{R}^n$, we have $B_{d_\infty}(x,\varepsilon) = (x_1-\varepsilon, x_1+\varepsilon) \times \dots \times (x_n-\varepsilon, x_n+\varepsilon)$

Hence the metric topology τ_{d_∞} coincides with the product topology.

Q: When do two metrics induce the same topology?

Definition 1: Two metrics d and d' on a set X are **equivalent** if there exist $c, c' \in (0,\infty)$ such that

$$c\, d(x,y) \le d'(x,y) \le c'\, d(x,y) \qquad \forall x,y \in X.$$

Note: 1) This just means that $\forall x \in X$ and $\forall \varepsilon > 0$,

$$B_{d'}(x, c\varepsilon) \subseteq B_d(x,\varepsilon) \subseteq B_{d'}(x, c'r).$$

2) We also get

$$\frac{1}{c'} d'(x,y) \le d(x,y) \le \frac{1}{c}\, d'(x,y)$$

3) This defines an equivalence relation on the set of metrics on X.

Proposition 2: If d and d' are equivalent metrics on a set X, then $\tau_d = \tau_{d'}$.

Proof: Let $c, c' \in (0,\infty)$ such that $c\,d(x,y) \le d'(x,y) \le c'\,d(x,y) \quad \forall x,y \in X$.

Thus $\forall \varepsilon > 0$ we have $B_{d'}(x, c\varepsilon) \subseteq B_d(x,\varepsilon) \subseteq B_{d'}(x, c'\varepsilon)$

We want to prove $\tau_d = \tau_{d'}$.

[$\subseteq$] Let $U \in \tau_d$. WTS $U \in \tau_{d'}$ $(= \{ V \subseteq X \mid \forall x \in V \ \exists \delta > 0$ such that $B_{d'}(x,\delta) \subseteq V \})$

Let $x \in U$. Since $U \in \tau_d$, $\exists\ \varepsilon > 0$ such that $B_d(x,\varepsilon) \subseteq U$. Since

$B_{d'}(x, c\varepsilon) \subseteq B_d(x,\varepsilon) \subseteq U$, we get $U \in \tau_{d'}$.

[$\supseteq$] Follows similarly, using that $B_d\left(x, \frac{1}{c'}\varepsilon\right) \subseteq B_{d'}(x,\varepsilon) \quad \forall x \in X \ \forall \varepsilon > 0$. □

Remark: The converse is not true in general. Given a metric space (X,d), one can show that $d'(x,y) = \frac{d(x,y)}{1 + d(x,y)}$ is also a metric on X which is <u>not</u> equivalent (in general) to d. However, they induce the same topology.

Example: On $X = \mathbb{R}^n$ we have metrics d_p for $p \in [1, \infty]$. One can show that $\forall x, y \in \mathbb{R}^n$,

$$d_\infty(x,y) \leq d_p(x,y) \leq n^{1/p} d_\infty(x,y)$$

So all these metrics are equivalent, hence induce the same topology.

Now back to continuity:

Proposition 3: Let (X, d_X) and (Y, d_Y) metric spaces and $f: X \to Y$ a function. Fix $a \in X$. The following are equivalent:

1) f is continuous at a.
2) $\forall \varepsilon > 0$, $\exists \delta > 0$ such that $f(B_{d_X}(a, \delta)) \subseteq B_{d_Y}(f(a), \varepsilon)$.

Proof: (1) $\Rightarrow$ (2) Assume f is continuous at a. Then, for each neighborhood U of $f(a) \in Y$, $\exists$ nghd V of $a \in X$ such that $f(V) \subseteq U$. Take arbitrary $\varepsilon > 0$. Since $B(f(a), \varepsilon)$ is a nghd of $f(a)$, $\exists V$ nghd of a such that $f(V) \subseteq B(f(a), \varepsilon)$. Since V nghd of a, $\exists$ open $V' \in \mathcal{T}_{d_X}$ with $a \in V' \subseteq V$. By definition of $\mathcal{T}_{d_X}$, this means there exists $\delta > 0$ such that $B_{d_X}(a, \delta) \subseteq V' \subseteq V$.

Hence $f(B(a,\delta)) \subseteq f(V') \subseteq f(V) \subseteq B(f(a),\varepsilon)$.

(2) $\Rightarrow$ (1) Let U nghbd of $f(a) \in Y$. WTS: $\exists$ V nghbd of a with $f(V) \subseteq U$.

Know: $\exists$ $U' \in T_{d_Y}$ with $f(a) \in U' \subseteq U$, hence $\exists$ $\varepsilon > 0$ with $B(f(a),\varepsilon) \subseteq U' \subseteq U$

Therefore, $\exists$ $\delta > 0$ such that $f(B(a,\delta)) \subseteq B(f(a),\varepsilon)$

We take $V = B(a,\delta)$, which is a neighborhood of $a \in X$ and satisfies

$$f(V) = f(B(a,\delta)) \subseteq B(f(a),\varepsilon) \subseteq U' \subseteq U,$$

hence $f(V) \subseteq U$ as desired $\square$